

Lecture 20: Practice Problems — Technical Case Studies

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Case Study 1: High-Frequency FinTech Bayesian Fraud & Anomaly Detection Pipeline

Context & Engineering Problem

A tier-1 payment processor handles 10,000,000 transactions daily. True fraudulent transactions are rare, with an established historical base rate:

$$P(\text{Fraud}) = 0.002 \quad (0.2\%)$$

The risk engineering team deploys two sequential fraud detection heuristic signals:

1. **Signal 1 (S_1 - Device Fingerprint Anomaly):**

- Sensitivity (True Positive Rate): $P(S_1 | \text{Fraud}) = 0.85$
- False Positive Rate: $P(S_1 | \text{Legitimate}) = 0.04$

2. **Signal 2 (S_2 - Impossible Geo-Velocity Movement):**

- Sensitivity (True Positive Rate): $P(S_2 | \text{Fraud}) = 0.90$
- False Positive Rate: $P(S_2 | \text{Legitimate}) = 0.03$

Assume that conditional on transaction class (Fraud vs. Legitimate), signals S_1 and S_2 are conditionally independent.

Mathematical & Implementation Tasks

1. **Single-Signal Bayesian Update:** Calculate the posterior probability of fraud given that only Signal 1 triggers ($P(\text{Fraud} \mid S_1)$). Explain why this posterior is surprisingly low despite high sensitivity (demonstrating the Base Rate Fallacy).
2. **Sequential Bayesian Updating:** Using the posterior probability from Task 1 as the new prior probability, calculate the updated posterior probability of fraud after Signal 2 also triggers ($P(\text{Fraud} \mid S_1, S_2)$).
3. **Simultaneous Verification:** Prove mathematically that sequential updating yields the exact same posterior as the joint simultaneous likelihood formulation:

$$P(\text{Fraud} \mid S_1, S_2) = \frac{P(S_1 \mid \text{Fraud})P(S_2 \mid \text{Fraud})P(\text{Fraud})}{P(S_1, S_2)}$$

4. **Python Pipeline:** Implement a reusable Python function that accepts any list of sequential binary diagnostic test results and returns the step-by-step posterior belief vector.

Case Study 2: Bioinformatics Clinical Drug Efficacy via Binomial Testing

Context & Engineering Problem

A biotechnology firm is conducting a Phase II clinical oncology trial for a novel immunotherapy agent across $n = 50$ terminal cancer patients.

- **Historical Standard-of-Care (SOC) Remission Rate:** $p_0 = 0.20$ (20% historical response rate).
- **Trial Outcome:** At the conclusion of the trial, $k = 18$ patients achieve verified complete tumor remission.

Regulatory agencies require formal statistical proof that the new compound is significantly superior to standard-of-care before granting Phase III progression.

Mathematical & Implementation Tasks

1. **Hypothesis Formulation:** Formulate the formal null (H_0) and alternative (H_1) statistical hypotheses for a one-tailed efficacy trial.

2. **Exact Binomial Tail Probability (p-value):** Under null hypothesis $H_0 : X \sim B(50, 0.20)$, derive and compute the exact one-tailed p-value:

$$p\text{-value} = P(X \geq 18 \mid n = 50, p_0 = 0.20) = \sum_{j=18}^{50} \binom{50}{j} (0.20)^j (0.80)^{50-j}$$

3. **Gaussian (Normal) Approximation with Continuity Correction:** Compute the expected mean $\mu = np_0$ and standard deviation $\sigma = \sqrt{np_0(1 - p_0)}$. Using the Yates continuity correction ($k - 0.5 = 17.5$), compute the standardized Z-score:

$$Z = \frac{(k - 0.5) - np_0}{\sqrt{np_0(1 - p_0)}}$$

Evaluate the Gaussian p-value $P(Z \geq z)$ and assess percentage error relative to the exact Binomial result. 4. **Statistical Decision:** Given standard significance threshold $\alpha = 0.01$, does the compound warrant clinical Phase III approval?

Case Study 3: Autonomous Vehicle LiDAR Sensor Calibration & Gaussian Noise Normalization

Context & Engineering Problem

An autonomous vehicle navigation subsystem uses a pulsed LiDAR sensor to estimate distance to roadside obstacles. Due to atmospheric dust, temperature fluctuation, and optical refraction, sensor distance readings exhibit zero-mean additive Gaussian noise:

$$X_{\text{meas}} = d_{\text{true}} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \quad \sigma = 0.15 \text{ meters}$$

The vehicle's collision avoidance system triggers three warning states:

- **Green (Normal):** Error within 1 standard deviation ($|\epsilon| \leq \sigma = 0.15 \text{ m}$)
- **Yellow (Advisory):** Error between 1 and 2 standard deviations ($0.15 \text{ m} < |\epsilon| \leq 0.30 \text{ m}$)

- **Red (Critical Safety Violation):** Error exceeds 3 standard deviations ($|\epsilon| > 0.45 \text{ m}$)

Mathematical & Implementation Tasks

1. **68-95-99.7 Empirical Integration:** Calculate the exact theoretical probability of the sensor operating in each state (Green, Yellow, and Red).
2. **Z-Score Standardization & Probability Density Integration:** Derive the probability that measurement error exceeds safety cutoff $\delta = 0.25 \text{ meters}$ using the Standard Normal CDF $\Phi(z)$:

$$P(|\epsilon| > 0.25) = 2 \cdot \left[1 - \Phi\left(\frac{0.25}{\sigma}\right) \right]$$

3. **Monte Carlo Calibration Simulation:** Simulate $N = 1,000,000$ synthetic LiDAR sensor pings. Compute sample mean, sample variance, empirical fraction of critical violations, and compare with theoretical predictions.